

Introducción al uso de registros nacionales de salud para investigación

Daniel R. Witte

Contents

Bienvenida	7
Datos clave	7
Público	7
Preparación	7
Re-utilización y licencia	7
Como se construyó este taller	7
.	7

I

Resumen

1	Temario	11
1.1	Objetivos de aprendizaje	11
2	¿Este taller es para mí?	13
3	Horario	15
3.1	10 de septiembre 2026	15
4	Learning design	17
4.1	Clear learning goals and a strong narrative	17
4.2	“Code-alongs” as the main activity	17
4.3	Retrieval practice with hands-on exercises	17
4.4	Linking new information with existing knowledge with discussion activities	17
4.5	Minimal use of lectures	17
4.6	Frequent breaks and variation in activities	17
4.7	Easily accessible material that is online, always	18
4.8	A safe and supportive environment	18
4.9	Using origami hats to ask for help	18
4.10	Being involved in the learning process	18

II

Pre-taller

5	Tareas preparatorias antes del taller	21
6	Instalación de los programas	23
6.1	Windows	23
6.2	MacOS	23
6.3	Linux (Ubuntu)	23

7	Instalación de los paquetes R necesarios	25
8	Code of conduct	27
8.1	Expected behavior	27
8.2	Unacceptable behavior	27
8.3	Consequences of unacceptable behavior	28

III

Sesiones

9	Introducción al taller	31
9.1	Tema de esta presentación	31
9.1.1	Objetivos de aprendizaje	31
10	El potencial de los registros	33
10.1	Objetivos de aprendizaje	33
10.2	TODO: Section	33
10.3	:technologist: Exercise: Title	33
10.4	:speech_balloon: Discussion activity: Title	33
10.5	:book: Reading task: Title	33
10.6	Summary	33
10.7	Survey	33
11	Partiendo de datos crudos	35
11.1	Objetivos de aprendizaje	35
11.2	Datos crudos	35
11.3	Datos analizables	35
11.4	Running Code	35
12	TODO: Lesson title here	37
12.1	Learning objectives	37
12.2	TODO: Section	37
12.3	:technologist: Exercise: Title	37
12.4	:speech_balloon: Discussion activity: Title	37
12.5	:book: Reading task: Title	37
12.6	Summary	37
12.7	Survey	37
13	What next	39
	Appendices	41
A	For teachers	43
	References	45

B	Contributing	47
B.1	:bug: Issues and bugs	47
B.2	:pencil2: Adding or modifying content	47
B.3	:file_folder: Support and infrastructure files	47
C	Creative Commons Attribution 4.0 International	49
C.1	Creative Commons Attribution 4.0 International Public License	49
C.1.1	Section 1 – Definitions.	49
C.1.2	Section 2 – Scope.	50
C.1.3	Section 3 – License Conditions.	51
C.1.4	Section 4 – Sui Generis Database Rights.	51
C.1.5	Section 5 – Disclaimer of Warranties and Limitation of Liability.	52
C.1.6	Section 6 – Term and Termination.	52
C.1.7	Section 7 – Other Terms and Conditions.	52
C.1.8	Section 8 – Interpretation.	52

Bienvenida



Bienvenidos al curso “Introducción al uso de registros nacionales de salud para investigación”. Este curso breve tiene como objetivo introducir los conceptos centrales del uso de registros de salud para investigación epidemiológica. El curso está enfocado en las enfermedades crónicas, especialmente en la diabetes pero los principios generales pueden aplicarse en otros contextos.

El curso se organiza por primera vez en 2026 en colaboración entre el [Grupo Crónicas](#), el [NCD Policy Lab](#) y [Steno Diabetes Center Aarhus](#).

Esta página web contiene todo el material del taller, incluyendo presentaciones y ejercicios. Está estructurado como un libro, con un “capítulo” para cada sesión. Haremos uso intensivo del internet durante el taller, donde las sesiones interactivas de programación en vivo siguen casi idénticamente los pasos en ésta página.

Datos clave



10 de septiembre 2026



9:00 - 12:00



Universidad Peruana Cayetano Heredia, sede Miraflores, Lima, Perú

Público

El curso está diseñado como taller para investigadores jóvenes y estudiantes con conocimientos en análisis de datos interesados en conocer metodologías de análisis con registros nacionales de salud.

Preparación

Para agilizar las sesiones el día del taller, les pedimos que completen las [tareas preparatorias](#) antes de llegar a la clase.

Re-utilización y licencia

El taller se publica con una licencia [Licencia Internacional Pública de Atribución/Reconocimiento 4.0 de Creative Commons](#). Quiere decir que los materiales pueden ser utilizados, re-utilizados y adaptados, con tal que cualquier nueva versión o reproducción parcial o total contenga reconocimiento y atribución a éste material como fuente.

Como se construyó este taller

Los materiales de este taller fueron creados usando el formato [Quarto](#), [Git](#) y [GitHub](#) para la gestión del repositorio y Github Pages para crear y alojar la página web.



Resumen

1	Temario	11
1.1	Objetivos de aprendizaje	11
2	¿Este taller es para mi?	13
3	Horario	15
3.1	10 de septiembre 2026	15
4	Learning design	17
4.1	Clear learning goals and a strong narrative	17
4.2	“Code-alongs” as the main activity	17
4.3	Retrieval practice with hands-on exercises	17
4.4	Linking new information with existing knowledge with discussion activities	17
4.5	Minimal use of lectures	17
4.6	Frequent breaks and variation in activities	17
4.7	Easily accessible material that is online, always .	18
4.8	A safe and supportive environment	18
4.9	Using origami hats to ask for help	18
4.10	Being involved in the learning process	18

1. Temario

Este taller dura 3 horas y consiste de las siguientes sesiones:

Consulta también el [horario](#)

- [Introducción al taller](#)
- [Potencial de los registros](#)
- [Partiendo de los datos crudos](#)
- [Un análisis sencillo](#)
- [Cierre](#)

1.1 Objetivos de aprendizaje

Hemos diseñado el taller para enseñarte lo siguiente:

1. Primer objetivo

2. ¿Este taller es para mí?

El curso está diseñado como taller para investigadores jóvenes y estudiantes con conocimientos en análisis de datos interesados en conocer metodologías de análisis con registros nacionales de salud.

Para ayudar a manejar las expectativas y para enfocar el desarrollo de los materiales, asumiremos cierta experiencia previa de los participantes de este taller. Esperamos que hayas estudiado, trabajado o tengas alguna experiencia con:

- Conceptos epidemiológicos básicos como prevalencia, determinantes, desenlaces, diseño de cohortes abiertas y cerradas, tiempo de seguimiento, censura, medidas de incidencia, medidas de riesgo, medidas de asociación, sesgos, confusión.
- Manejo de datos cuantitativos, conceptos básicos de la ciencia de datos como formatos *long* y *wide*, fusión de datos determinística y probabilística.
- Análisis de datos con programas estadísticos como R.

While we have these assumptions to help focus the content of the workshop, if you have an interest in the workshop but don't fit any of the assumptions, *you are still welcome to attend!* We welcome everyone until capacity has been reached.

In addition to the assumptions above, the workshop also has a fairly focused scope, which may also help you decide if this workshop is for you:

- TODO: List of concrete tasks we will do and not do.

3. Horario

Este taller contiene presentaciones, ejercicios participatorios con programación en vivo utilizando datos simulados, discusiones y trabajo en grupos.

3.1 10 de septiembre 2026

Horario	Módulo	Detalle	Expositor
09:00 - 09:10	Bienvenida	Presentación del docente y de las instituciones organizadoras.	
09:10 - 10:00	El potencial de los registros	Qué son los registros nacionales y poblacionales, y qué los hace tan valiosos. El modelo nórdico como ejemplo. Qué preguntas de investigación se vuelven posibles con estos datos que de otro modo serían muy difíciles. Fortalezas y limitaciones en lenguaje sencillo. Cierre del bloque contrastando con el contexto latinoamericano/peruano.	DW
10:00 - 10:15	Break		
10:15 - 11:00	De los datos crudos a una pregunta de investigación	Cómo se pasa de una base administrativa gigante a un conjunto de datos analizable. Ideas básicas: definir una población, elegir variables a partir de códigos, construir un seguimiento en el tiempo.	DW
11:00 - 11:50	Manos a la obra: un análisis sencillo	Un ejercicio guiado y simple con datos sintéticos o de ejemplo, donde todos siguen el mismo paso a paso: cargar los datos, mirar la estructura, y correr un análisis básico.	DW
11:50 - 12:00	Cierre. Ideas y discusión	Qué se necesitaría para hacer esto en nuestra	Todos

Horario	Módulo	Detalle	Expositor
		región/país y ronda de preguntas.	

4. Learning design

This workshop is designed using the science of learning. Learning is when new content gets incorporated into long-term storage in the brain in a way that it can be later retrieved. Some of the best ways to enable learning is to do “active learning” by engaging with the material, practicing retrieving what was just covered, repeatedly being exposed to the content, linking the new information with what you already know, and explaining it to someone else.

We use a variety of activities to target the different “pathways” of learning. A big part of learning is being engaged in the *process* of learning, not just learning the content itself. So to help you be engaged in your own learning, we want to involve you in the reasons *why* we do the things we do during this workshop.

Throughout the workshop, we use these practices to support learning.

4.1 Clear learning goals and a strong narrative

The learning goals and the narrative are tightly connected in this workshop. Learning is easier when you have a better idea of where you are going, where you are relative to the end goal, and what to expect along the way.

4.2 “Code-alongs” as the main activity

“Code-alongs” is when the teacher types on their computer that is on the projector and explains what it means as you also type along with them. The connection between using the hands and doing something tangible helps reinforce learning. It also helps the teachers to go slower as we have to both explain and type things out. This helps you learn better. “Less is more” works very well with learning.

4.3 Retrieval practice with hands-on exercises

Hands-on exercises reinforce what was just taught through retrieval practice and repetition. It gives you a chance to engage with the material and the space to try it out on your own. This helps you form a better sense of how to do it yourself.

4.4 Linking new information with existing knowledge with discussion activities

Discussion activities encourage linking what you already know and do in your work with the new information. Because humans are social beings, we also tend to learn better when there is some social aspect to it. For example, explaining something to someone else is one of the best ways to learn, as nicely expressed in the quote “while we teach, we learn” from the Roman philosopher Seneca.

4.5 Minimal use of lectures

Lectures, while extremely common, are forms of information sharing with the lowest rates of learning. In this workshop, we use reading tasks instead, so you can go at your pace, re-read things you may not understand, and take notes, all of which support learning. To help reinforce what you just read, the teacher will briefly summarise and emphasise key points after reading tasks, to repeat the main points and target another part of the brain (the auditory pathway).

4.6 Frequent breaks and variation in activities

Breaks are how your brain can recharge and refocus. The brain is only able to effectively focus on one thing at a time in 30-50 minute blocks of time. That’s why we mix together code-alongs, exercises, discussions,

and reading to give your brain some variation. Having an explicit break, even 5 minutes, is enough for the brain to better process and store what was covered.

4.7 Easily accessible material that is online, always

All the material is online on this website, so that it is easy for you to go back to after the workshop. This builds the repetition and retrieval practice part of learning. Our brains also get easily distracted, which is completely normal. So if you get distracted during the workshop, you can simply go to the material and get caught up with where we are. We also make sure to follow the material on the website closely, so that it is easier for you to follow along.

4.8 A safe and supportive environment

Mental safety in a supportive environment is critical for effective learning. It puts the brain into a more relaxed state that allows for optimal processing and ultimately learning. That's why we try to explicitly create that space, for example with the use of our Code of Conduct.

4.9 Using origami hats to ask for help

Asking for help can feel difficult, especially when you are learning and are unsure about something. To make it easier and feel safer, we use origami hats (or stickies) to show how everyone is doing. If you need help, you put up the "help" hat (usually orange) and a helper will come to you as soon as possible. If not, you can put up a "all good" hat (usually blue) to let us know you are following along. Since we often have several helpers, you usually won't wait long. This simple system lets you ask for help without interrupting the teacher or drawing attention, so you can keep following along while you wait. Our goal is to make getting help easier and more comfortable.

4.10 Being involved in the learning process

Now you know a bit more about the thoughts behind how we've designed this workshop to really try to support and enable your own learning! :tada: :grin:

II

Pre-taller

5	Tareas preparatorias antes del taller .	21
6	Instalación de los programas	23
6.1	Windows	23
6.2	MacOS	23
6.3	Linux (Ubuntu)	23
7	Instalación de los paquetes R necesarios	25
8	Code of conduct	27
8.1	Expected behavior	27
8.2	Unacceptable behavior	27
8.3	Consequences of unacceptable behavior	28

5. Tareas preparatorias antes del taller

Para agilizar el taller es importante que todos los participantes completen las siguientes tareas preparatorias:

1. Instala los programas R, RStudio, Quarto, y Git en sus versiones mas actuales de acuerdo con [?@sec-installing-programs](#).
2. Instala los paquetes R necesarios.
3. Lee el temario (Chapter 1.).

Cada sección contiene detalles sobre cómo completar cada tarea.

Tip

Puedes pasar a la siguiente sección haciendo click en la flechita en la parte inferior de cada página :

6. Instalación de los programas

Para poder participar en el taller, es importante que tengas los siguientes programas instalados en tu computadora. Es posible que ya hayas instalado algunos de ellos en el pasado. En ese caso por favor verifica que tengas instalada una versión mas reciente que la indicada en las instrucciones. Si no tendrás que actualizar las instalaciones.

6.1 Windows

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`.
2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”.
3. **Git**: Escoge el enlace “Click here to download”. Usaremos Git durante algunas partes del taller. Al instalar Git, te pedirá seleccionar un “Text Editor” y aunque no usaremos esa funcion en el taller es necesario proporcionar el dato para la instalación de Git, así que lo mas fácil es escoger Notepad.

6.2 MacOS

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`. Si usas [Homebrew](#) (recomendado en general pero), la instalación de R is muy fácil. Solo hay que dar el siguiente comando en el Terminal de tu computadora:

```
brew install --cask r
```

2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”. Con Homebrew:

```
brew install --cask rstudio
```

3. **Git**: Usaremos Git durante algunas partes del taller. MacOS viene con Git pre-instalado. Para instalarlo con Homebrew:

```
brew install git
```

6.3 Linux (Ubuntu)

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`.

```
sudo apt -y install r-base
```

2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”.
3. **Git**: Usaremos Git durante algunas partes del taller.

```
sudo apt install git
```

Todos estos programas son necesarios para el taller, Git incluido. Git es usado para la gestión formal de versiones de archivos. Lo usamos porque tiene mucha documentación muy buena. Puedes aprender mas en éste libro en línea [Happy Git with R](#). Especialmente la sección “Why Git” explica bien por qué lo usamos. Usuarios de Windows a veces encuentran dificultades durante la instalación de Git. La siguiente guia ofrece pautas y ayuda [Installing Git for Windows](#).

7. Instalación de los paquetes R necesarios

8. Code of conduct

Learning happens when we feel safe and supported. Since we want you to learn during this workshop, we want you to feel safe and supported. We also want all learners to have an enjoyable and fulfilling experience while participating in this workshop. So we expect all learners to show respect, kindness, and courtesy to other learners throughout the whole workshop and to help create a positive and safe learning environment.

While most people and (sub-)cultures have an implicit, assumed set of practices and expectations for behaviour, we believe strongly in effective and clear communication. This includes making what is normally implicit expectations to instead be explicit expectations.

Teachers and helpers will uphold and embody this code in all spaces and activities of the workshop, and we expect you as the learner to do the same. In fact, by participating in this workshop, you indicate that you agree to follow these explicit expectations. We have this code of conduct to communicate to you that we really value and want to nurture a space that encourages and enhances learning. Otherwise, why would we have a workshop if you can't or aren't able to effectively learn anything from it! :zany_face:

8.1 Expected behavior

We want a welcoming, supportive, safe, and fun environment not only for ourselves as teachers and helpers, but also for all learners, regardless of background and identity. This means that we do not tolerate behaviour that is disrespectful to any of our teachers, helpers, or learners (see below for unacceptable behaviour).

We encourage the following kinds of behaviours in all interactions during this workshop that will foster a positive and professional learning environment:

- Show kindness, courtesy, and respect towards other workshop members.
- All interactions should be professional regardless of platform: either online or in-person.
- Be aware of your words by using welcoming and inclusive language. Words matter, a lot!
- Be respectful to different viewpoints and experiences.
- Gracefully accept constructive criticism.
- Focus on what is best for everyone in the workshop.

8.2 Unacceptable behavior

While we assume you will not do these behaviours, we want to clearly communicate what unacceptable behaviour is. It is any behaviour, as actions or words, that excludes, intimidates, or causes discomfort to others. We do not tolerate discrimination or harassment based on characteristics that include, though are not limited to, gender identity and expression, sexual orientation, disability, physical appearance, body size, citizenship, nationality, ethnic or social origin, pregnancy, familial status, veteran status, genetic information, religion or belief (or lack thereof), membership of a national minority, property, age, education, socio-economic status, technical choices, and experience level. We of course also don't tolerate verbally or physically violent behaviour towards others.

Examples of unacceptable behavior includes:

- Written or verbal comments that exclude people on the basis of membership of any specific group.
- The display of sexual or violent images.
- Unwelcome sexual attention, advances, and/or physical contact.
- Sustained disruption of workshop sessions or communications.
- Insults or put downs.
- Sexist, racist, homophobic, transphobic, ableist, or exclusionary jokes.

8.3 Consequences of unacceptable behavior

Learners who are asked, either in person or online, to stop any inappropriate behavior are expected to comply immediately. If a learner engages in behaviour that violates this code of conduct, the organizers, teachers, or helpers may take any action they deem appropriate, including warning the offender or removing them from the workshop.

If you feel in any way unsafe or mistreated, you may contact us privately—all communication is confidential.

Thank you for helping make this a welcoming, friendly workshop for all :smile:

Note

Acknowledgements: Large portions of this code of conduct were taken and modified from other codes of conduct, specifically [Software Carpentry](#) and [UofTCoders](#).

III

Sesiones

9	Introducción al taller	31
9.1	Tema de esta presentación	31
10	El potencial de los registros	33
10.1	Objetivos de aprendizaje	33
10.2	TODO: Section	33
10.3	:technologist: Exercise: Title	33
10.4	:speech_balloon: Discussion activity: Title	33
10.5	:book: Reading task: Title	33
10.6	Summary	33
10.7	Survey	33
11	Partiendo de datos crudos	35
11.1	Objetivos de aprendizaje	35
11.2	Datos crudos	35
11.3	Datos analizables	35
11.4	Running Code	35
12	TODO: Lesson title here	37
12.1	Learning objectives	37
12.2	TODO: Section	37
12.3	:technologist: Exercise: Title	37
12.4	:speech_balloon: Discussion activity: Title	37
12.5	:book: Reading task: Title	37
12.6	Summary	37
12.7	Survey	37
13	What next	39

9. Introducción al taller

Diapositivas de la introducción general en línea

Nota para el profesor

Las diapositivas pueden contener notas que se pueden visualizar con la 'S' en el teclado

9.1 Tema de esta presentación

Esta presentación da una introducción general a los conceptos centrales del trabajo con registros sanitarios. Se dicta como primera sesión o como presentación pre-curso (en línea).

9.1.1 Objetivos de aprendizaje

Después de esta presentación los estudiantes tendrán los siguientes conocimientos:

- Qué es un registro sanitario; con cuales componentes debe contar
- A qué tipo de preguntas puede responder un registro
- Cuales son los tipos de usuarios de los datos de un registro sanitario
- A qué niveles existen datos / registros sanitarios y cuales son las necesidades de información a cada nivel

10. El potencial de los registros

En esta sesión repasaremos rápidamente algunas de las definiciones

10.1 Objetivos de aprendizaje

El objetivo de esta sesión es que el estudiante conozca y pueda explicar:

- Qué son los registros sanitarios poblacionales, con enfoque en enfermedades crónicas
- Cómo se construyen y mantienen
- Qué los hace tan valiosos
- Cual es la experiencia de los países nórdicos y de Brasil

Tomando el modelo nórdico como ejemplo discutiremos qué tipo de preguntas de investigación se vuelven posibles con estos datos que de otro modo serían muy difíciles. Discutiremos fortalezas, limitaciones y retos. Cerraremos el bloque contrastando con el contexto latinoamericano/peruano.

10.2 TODO: Section

i Reading task: ~4 minutes

TODO: Reading text

10.3 :technologist: Exercise: Title

Time: ~NUM minutes.

Text

10.4 :speech_balloon: Discussion activity: Title

Time: ~Num minutes.

Text

10.5 :book: Reading task: Title

Time: ~Num minutes.

Text

10.6 Summary

- TODO: List of summary items

10.7 Survey

Feedback survey! :tada:

11. Partiendo de datos crudos

11.1 Objetivos de aprendizaje

En esta sesión vamos a investigar la estructura de las bases de datos del registro de diabetes de Dinamarca, basándonos en datos simulados. Esto quiere decir que todos los datos que veremos son generados aleatóreamente y no pertenecen a ninguna persona real.

Usaremos el paquete “osdc” (Open Source Diabetes Classifier). Antes de iniciar esta sesión debes haber completado las tareas pre-taller.

El primer paso es verificar si está instalado el paquete osdc.

```
install.packages("osdc")
```

Podemos consultar la [documentación del paquete osdc](#) (en Inglés)

Para usar un paquete en una sesión debemos comenzar cargándolo

```
library(osdc)
```

El paquete contiene varias funciones, pero también documentación en forma de guías o viñetas. Podemos revisarlas dentro del ambiente R.

La primera nos da la documentación del algoritmo

```
vignette("algorithm")
```

El siguiente paso es revisar todas las fuentes de datos que son necesarias para crear el registro.

```
vignette("data-sources")
```

Como se puede ver, hay 5 registros, pero en algunos casos un solo registro tiene varias tablas temáticas que se acceden como bases separadas (Landspatientsregisterets) o separadas en el tiempo (Sygesikringsregisteret)

La segunda tabla nos muestra todas las variables de cada registro que son necesarias para construir el registro de diabetes.

11.2 Datos crudos

11.3 Datos analizables

11.4 Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

12. TODO: Lesson title here

12.1 Learning objectives

The learning objectives for this session are:

1. Objective one

12.2 TODO: Section

i Reading task: ~4 minutes

TODO: Reading text

12.3 :technologist: Exercise: Title

Time: ~NUM minutes.

Text

12.4 :speech_balloon: Discussion activity: Title

Time: ~Num minutes.

Text

12.5 :book: Reading task: Title

Time: ~Num minutes.

Text

12.6 Summary

- TODO: List of summary items

12.7 Survey

Feedback survey! :tada:

13. What next

Slides

Teacher note

The slides contain speaking notes that you can view by pressing 'S' on the keyboard.

Appendices

A. For teachers

A more general guide to teaching or helping is found on the [Guides](#) website. This page has information relevant specifically to this workshop.

- For created teams, assign one teacher or helper to each group to help them specifically during the final group project.

References

B. Contributing

Check out the [Contributing Guide](#) for more details on how to contribute. For guidelines on how to be a helper or teacher, check out the [For teachers](#) page. As always, make sure to adhere to the [Code of Conduct](#).

B.1 :bug: Issues and bugs

The easiest way to contribute is to report issues or bugs that you might find while using or reading the workshop material. You can do this by creating a new issue on our GitHub repository.

B.2 :pencil2: Adding or modifying content

To contribute to the workshop, you first need to install [uv](#) and [justfile](#). We use [uv](#) and [justfile](#) to manage our project, such as to run checks and build the workshop website. Both the [uv](#) and [justfile](#) websites have a more detailed guide on using [uv](#), but below are some simple instructions to get you started.

It's easiest to [install uv](#) first and then install [justfile](#) using [uv](#). Then, to install [justfile](#), run:

```
uv tool install rust-just
```

We keep all our development workflows in the [justfile](#), so you can explore it to see what commands are available. To see a list of commands available, run:

```
just
```

As you contribute, make sure your changes will pass our checks by opening a terminal so that the working directory is the root of this workshop's parent/root folder and running:

```
just run-all
```

When committing changes, please try to follow [Conventional Commits](#) as Git messages. Using this convention allows us to be able to automatically create a release based on the commit message by using [Cocogitto](#). If you don't use Conventional Commits when making a commit, we will revise the pull request title to follow that format. That's because we use squash merges when merging pull requests, so all other commits in the pull request will be squashed into one commit.

B.3 :file_folder: Support and infrastructure files

- [.copier-answers.yml](#): Contains the answers you gave when copying the project from the template. **You should not modify this file directly.**
- [.gitignore](#): This ignore file tells Git which files to not track. Unless you know what you are doing, it's best to not touch this file.
- [_quarto.yml](#): Quarto configuration file for the website, including settings for the website, such as the theme, navigation, and other options.
- [_metadata.yml](#): Quarto metadata file for the website, including information about the project, such as the titles and GitHub names.
- [.pre-commit-config.yaml](#): [Pre-commit](#) configuration file for managing and running checks before each commit.
- [.config/](#): Contains configuration files for various tools used in the project, such as:
 - [typos.toml](#): [typos](#) spell checker configuration file.
 - [rmdl.toml](#) and [panache.toml](#): [rmdl](#) and [Panache](#) configuration file for formatting Markdown files in the project.
- [.zenodo.json](#): Structured citation metadata for your project when archived on [Zenodo](#). This is used to add the metadata to Zenodo when a GitHub release has been uploaded to Zenodo.
- [justfile](#): [just](#) configuration file for scripting project tasks.

- `.editorconfig`: Editor configuration file for [EditorConfig](#) to maintain consistent coding styles across different editors and IDEs.
- `CONTRIBUTING.md`: Guidelines for contributing to the project.
- `.github/`: Contains GitHub-specific files, such as issue and pull request templates, workflows, [dependabot](#) configuration, pull request templates, and a [CODEOWNERS](#) file.

C. Creative Commons Attribution 4.0 International

Creative Commons Corporation (“Creative Commons”) is not a law firm and does not provide legal services or legal advice. Distribution of Creative Commons public licenses does not create a lawyer-client or other relationship. Creative Commons makes its licenses and related information available on an “as-is” basis. Creative Commons gives no warranties regarding its licenses, any material licensed under their terms and conditions, or any related information. Creative Commons disclaims all liability for damages resulting from their use to the fullest extent possible.

Using Creative Commons Public Licenses

Creative Commons public licenses provide a standard set of terms and conditions that creators and other rights holders may use to share original works of authorship and other material subject to copyright and certain other rights specified in the public license below. The following considerations are for informational purposes only, are not exhaustive, and do not form part of our licenses.

- **Considerations for licensors:** Our public licenses are intended for use by those authorized to give the public permission to use material in ways otherwise restricted by copyright and certain other rights. Our licenses are irrevocable. Licensors should read and understand the terms and conditions of the license they choose before applying it. Licensors should also secure all rights necessary before applying our licenses so that the public can reuse the material as expected. Licensors should clearly mark any material not subject to the license. This includes other CC-licensed material, or material used under an exception or limitation to copyright. [More considerations for licensors.](#)
- **Considerations for the public:** By using one of our public licenses, a licensor grants the public permission to use the licensed material under specified terms and conditions. If the licensor’s permission is not necessary for any reason—for example, because of any applicable exception or limitation to copyright—then that use is not regulated by the license. Our licenses grant only permissions under copyright and certain other rights that a licensor has authority to grant. Use of the licensed material may still be restricted for other reasons, including because others have copyright or other rights in the material. A licensor may make special requests, such as asking that all changes be marked or described. Although not required by our licenses, you are encouraged to respect those requests where reasonable. [More considerations for the public.](#)

C.1 Creative Commons Attribution 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree to be bound by the terms and conditions of this Creative Commons Attribution 4.0 International Public License (“Public License”). To the extent this Public License may be interpreted as a contract, You are granted the Licensed Rights in consideration of Your acceptance of these terms and conditions, and the Licensor grants You such rights in consideration of benefits the Licensor receives from making the Licensed Material available under these terms and conditions.

C.1.1 Section 1 – Definitions.

- a. **Adapted Material** means material subject to Copyright and Similar Rights that is derived from or based upon the Licensed Material and in which the Licensed Material is translated, altered, arranged, transformed, or otherwise modified in a manner requiring permission under the Copyright and Similar Rights held by the Licensor. For purposes of this Public License, where the Licensed Material is a musical work, performance, or sound recording, Adapted Material is always produced where the Licensed Material is synched in timed relation with a moving image.
- b. **Adapter’s License** means the license You apply to Your Copyright and Similar Rights in Your contributions to Adapted Material in accordance with the terms and conditions of this Public License.

- c. **Copyright and Similar Rights** means copyright and/or similar rights closely related to copyright including, without limitation, performance, broadcast, sound recording, and Sui Generis Database Rights, without regard to how the rights are labeled or categorized. For purposes of this Public License, the rights specified in Section 2(b)(1)-(2) are not Copyright and Similar Rights.
- d. **Effective Technological Measures** means those measures that, in the absence of proper authority, may not be circumvented under laws fulfilling obligations under Article 11 of the WIPO Copyright Treaty adopted on December 20, 1996, and/or similar international agreements.
- e. **Exceptions and Limitations** means fair use, fair dealing, and/or any other exception or limitation to Copyright and Similar Rights that applies to Your use of the Licensed Material.
- f. **Licensed Material** means the artistic or literary work, database, or other material to which the Licensor applied this Public License.
- g. **Licensed Rights** means the rights granted to You subject to the terms and conditions of this Public License, which are limited to all Copyright and Similar Rights that apply to Your use of the Licensed Material and that the Licensor has authority to license.
- h. **Licensor** means the individual(s) or entity(ies) granting rights under this Public License.
- i. **Share** means to provide material to the public by any means or process that requires permission under the Licensed Rights, such as reproduction, public display, public performance, distribution, dissemination, communication, or importation, and to make material available to the public including in ways that members of the public may access the material from a place and at a time individually chosen by them.
- j. **Sui Generis Database Rights** means rights other than copyright resulting from Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, as amended and/or succeeded, as well as other essentially equivalent rights anywhere in the world.
- k. **You** means the individual or entity exercising the Licensed Rights under this Public License. **Your** has a corresponding meaning.

C.1.2 Section 2 – Scope.

- a. **License grant.**
 - a. Subject to the terms and conditions of this Public License, the Licensor hereby grants You a worldwide, royalty-free, non-sublicensable, non-exclusive, irrevocable license to exercise the Licensed Rights in the Licensed Material to:
 - A. reproduce and Share the Licensed Material, in whole or in part; and
 - B. produce, reproduce, and Share Adapted Material.
 - b. **Exceptions and Limitations.** For the avoidance of doubt, where Exceptions and Limitations apply to Your use, this Public License does not apply, and You do not need to comply with its terms and conditions.
 - c. **Term.** The term of this Public License is specified in Section 6(a).
 - d. **Media and formats; technical modifications allowed.** The Licensor authorizes You to exercise the Licensed Rights in all media and formats whether now known or hereafter created, and to make technical modifications necessary to do so. The Licensor waives and/or agrees not to assert any right or authority to forbid You from making technical modifications necessary to exercise the Licensed Rights, including technical modifications necessary to circumvent Effective Technological Measures. For purposes of this Public License, simply making modifications authorized by this Section 2(a)(4) never produces Adapted Material.
 - e. **Downstream recipients.**
 - A. **Offer from the Licensor – Licensed Material.** Every recipient of the Licensed Material automatically receives an offer from the Licensor to exercise the Licensed Rights under the terms and conditions of this Public License.
 - B. **No downstream restrictions.** You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, the Licensed Material if doing so restricts exercise of the Licensed Rights by any recipient of the Licensed Material.

f. **No endorsement.** Nothing in this Public License constitutes or may be construed as permission to assert or imply that You are, or that Your use of the Licensed Material is, connected with, or sponsored, endorsed, or granted official status by, the Licensor or others designated to receive attribution as provided in Section 3(a)(1)(A)(i).

b. **Other rights.**

- a. Moral rights, such as the right of integrity, are not licensed under this Public License, nor are publicity, privacy, and/or other similar personality rights; however, to the extent possible, the Licensor waives and/or agrees not to assert any such rights held by the Licensor to the limited extent necessary to allow You to exercise the Licensed Rights, but not otherwise.
- b. Patent and trademark rights are not licensed under this Public License.
- c. To the extent possible, the Licensor waives any right to collect royalties from You for the exercise of the Licensed Rights, whether directly or through a collecting society under any voluntary or waivable statutory or compulsory licensing scheme. In all other cases the Licensor expressly reserves any right to collect such royalties.

C.1.3 Section 3 – License Conditions.

Your exercise of the Licensed Rights is expressly made subject to the following conditions.

a. **Attribution.**

- a. If You Share the Licensed Material (including in modified form), You must:
 - A. retain the following if it is supplied by the Licensor with the Licensed Material:
 - i. identification of the creator(s) of the Licensed Material and any others designated to receive attribution, in any reasonable manner requested by the Licensor (including by pseudonym if designated);
 - ii. a copyright notice;
 - iii. a notice that refers to this Public License;
 - iv. a notice that refers to the disclaimer of warranties;
 - v. a URI or hyperlink to the Licensed Material to the extent reasonably practicable;
 - B. indicate if You modified the Licensed Material and retain an indication of any previous modifications; and
 - C. indicate the Licensed Material is licensed under this Public License, and include the text of, or the URI or hyperlink to, this Public License.
- b. You may satisfy the conditions in Section 3(a)(1) in any reasonable manner based on the medium, means, and context in which You Share the Licensed Material. For example, it may be reasonable to satisfy the conditions by providing a URI or hyperlink to a resource that includes the required information.
- c. If requested by the Licensor, You must remove any of the information required by Section 3(a)(1)(A) to the extent reasonably practicable.
- d. If You Share Adapted Material You produce, the Adapter's License You apply must not prevent recipients of the Adapted Material from complying with this Public License.

C.1.4 Section 4 – Sui Generis Database Rights.

Where the Licensed Rights include Sui Generis Database Rights that apply to Your use of the Licensed Material:

- a. for the avoidance of doubt, Section 2(a)(1) grants You the right to extract, reuse, reproduce, and Share all or a substantial portion of the contents of the database;
- b. if You include all or a substantial portion of the database contents in a database in which You have Sui Generis Database Rights, then the database in which You have Sui Generis Database Rights (but not its individual contents) is Adapted Material; and
- c. You must comply with the conditions in Section 3(a) if You Share all or a substantial portion of the contents of the database.

For the avoidance of doubt, this Section 4 supplements and does not replace Your obligations under this Public License where the Licensed Rights include other Copyright and Similar Rights.

C.1.5 Section 5 – Disclaimer of Warranties and Limitation of Liability.

- a. Unless otherwise separately undertaken by the Licensor, to the extent possible, the Licensor offers the Licensed Material as-is and as-available, and makes no representations or warranties of any kind concerning the Licensed Material, whether express, implied, statutory, or other. This includes, without limitation, warranties of title, merchantability, fitness for a particular purpose, non-infringement, absence of latent or other defects, accuracy, or the presence or absence of errors, whether or not known or discoverable. Where disclaimers of warranties are not allowed in full or in part, this disclaimer may not apply to You.
- b. To the extent possible, in no event will the Licensor be liable to You on any legal theory (including, without limitation, negligence) or otherwise for any direct, special, indirect, incidental, consequential, punitive, exemplary, or other losses, costs, expenses, or damages arising out of this Public License or use of the Licensed Material, even if the Licensor has been advised of the possibility of such losses, costs, expenses, or damages. Where a limitation of liability is not allowed in full or in part, this limitation may not apply to You.
- c. The disclaimer of warranties and limitation of liability provided above shall be interpreted in a manner that, to the extent possible, most closely approximates an absolute disclaimer and waiver of all liability.

C.1.6 Section 6 – Term and Termination.

- a. This Public License applies for the term of the Copyright and Similar Rights licensed here. However, if You fail to comply with this Public License, then Your rights under this Public License terminate automatically.
- b. Where Your right to use the Licensed Material has terminated under Section 6(a), it reinstates:
 - a. automatically as of the date the violation is cured, provided it is cured within 30 days of Your discovery of the violation; or
 - b. upon express reinstatement by the Licensor.

For the avoidance of doubt, this Section 6(b) does not affect any right the Licensor may have to seek remedies for Your violations of this Public License.

- c. For the avoidance of doubt, the Licensor may also offer the Licensed Material under separate terms or conditions or stop distributing the Licensed Material at any time; however, doing so will not terminate this Public License.
- d. Sections 1, 5, 6, 7, and 8 survive termination of this Public License.

C.1.7 Section 7 – Other Terms and Conditions.

- a. The Licensor shall not be bound by any additional or different terms or conditions communicated by You unless expressly agreed.
- b. Any arrangements, understandings, or agreements regarding the Licensed Material not stated herein are separate from and independent of the terms and conditions of this Public License.

C.1.8 Section 8 – Interpretation.

- a. For the avoidance of doubt, this Public License does not, and shall not be interpreted to, reduce, limit, restrict, or impose conditions on any use of the Licensed Material that could lawfully be made without permission under this Public License.
- b. To the extent possible, if any provision of this Public License is deemed unenforceable, it shall be automatically reformed to the minimum extent necessary to make it enforceable. If the provision cannot be reformed, it shall be severed from this Public License without affecting the enforceability of the remaining terms and conditions.
- c. No term or condition of this Public License will be waived and no failure to comply consented to unless expressly agreed to by the Licensor.

- d. Nothing in this Public License constitutes or may be interpreted as a limitation upon, or waiver of, any privileges and immunities that apply to the Licensor or You, including from the legal processes of any jurisdiction or authority.

Creative Commons is not a party to its public licenses. Notwithstanding, Creative Commons may elect to apply one of its public licenses to material it publishes and in those instances will be considered the “Licensor.” Except for the limited purpose of indicating that material is shared under a Creative Commons public license or as otherwise permitted by the Creative Commons policies published at creativecommons.org/policies, Creative Commons does not authorize the use of the trademark “Creative Commons” or any other trademark or logo of Creative Commons without its prior written consent including, without limitation, in connection with any unauthorized modifications to any of its public licenses or any other arrangements, understandings, or agreements concerning use of licensed material. For the avoidance of doubt, this paragraph does not form part of the public licenses.

Creative Commons may be contacted at creativecommons.org